# Yash

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## Academic Qualifications

| Year | Degree/Certificate | Institute | CPI/% |
|---|---|---|---|
| 2024-Present | B.Tech | Indian Institute of Technology Kanpur | 6.2/10.0 |
| 2023 | CBSE(XII) | SVP Shikshan Awan Anusandhan Sansthan | 87.6% |
| 2021 | CBSE(X) | HHS New Eraa SSS, Sant Nagar, Sirsa | 86.0% |

## Work Experience

**Automations & Growth Operations Intern** | The Headlines Studio *(Apr'26–Jun'26)*

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| **Objective** | • Automated **end-to-end outreach** and **CRM** tracking to reduce manual lead-generation effort at scale robustly<br>• Standardized **domain & inbox warmup** protocols to protect **sender reputation** & delivery across campaigns |
| **Approach** | • Built scalable **email sequencing** pipelines integrated with **CRM** and tracker tools to power outreach workflows<br>• Connected third-party **automation platforms** to internal systems, replacing manual data entry steps entirely<br>• Documented and rigorously **tested workflows** thoroughly before handoff to studio's operating team members |
| **Result** | • Delivered fully **automated outreach workflows**, successfully embraced broadly, across **business** operations<br>• Improved **cross-functional efficiency** significantly, reducing manual overhead in daily **production** operations<br>• Drove continuous **process improvements** to expand output and accelerate overall studio trajectory gradually |

## Research Experience

**Computational Photography & AI Synthesis** | Prof. Tushar Sandhan | IIT Kanpur | *(Jul'25–Nov'25)*

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| **Objective** | • Engineered a **quantitative inverse-rendering** pipeline using OLAT, HDRI, **10+ lighting configs** for model<br>• Recruited and coordinated **80–100 volunteers** to build a **custom relighting dataset** for more precise model |
| **Approach** | • Trained a **3D U-Net (ResNet-34)** model for **dense regression analysis** on multi-source financial datasets<br>• Performed a precise **camera calibration** using **71 chessboard images** to correct financial distortion metrics |
| **Result** | • Automated **3D synthetic dataset generation** across **50+ configurations** for scalable financial data coverage<br>• Designed a **multi-signal modeling engine** with **depth-aware compositing** for precise, data-driven outcomes |

## Key Projects

**Self-Optimal Clustering (SOC) Technique** | EE656, Prof. Nishchal K. Verma, IIT Kanpur | *(Jun'26)*

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| **Objective** | • Reimplemented the **Self-Optimal Clustering (SOC)** from an **IEEE** paper entirely in **MATLAB** from scratch<br>• Benchmarked SOC against five methods: **K-Means, FCM, K-Medoid, IMC-1**, **IMC-2** on real **RGB** images |
| **Approach** | • Engineered **silhouette-driven adaptive threshold** via **Lagrange interpolation**, tuning cluster radii finely<br>• Vectorized **chunked potential evaluation** to fully process **62,500-pixel** images in **2GB**, bypass 30GB matrix |
| **Results** | • Automated **k=2–6** cluster-count selection via **GSI** maximization, achieving best **GSI, PI, SI** on cluster scene<br>• Built a pipeline computing **four** validation indices (**GSI, PI, SI, DI**) from first principles, across all six models |

**Smart Multimeter Simulation** | Electrical Engineering Association, IIT Kanpur *(Dec'25–Jan'26)*

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| **Objective** | • Developed **the** high-fidelity simulation of an industrial-grade **digital multimeter** for electrical measurements<br>• Designed the system architecture to support seamless real-time **voltage, current, and resistance** monitoring |
| **Approach** | • Engineered an adaptive **auto-ranging algorithm** spanning over $10^5$ measurement levels with high reliability<br>• Integrated **OTG-enabled telemetry** ensuring smooth real-time data acquisition, monitoring and visualization |
| **Results** | • Achieved nearly **98% measurement accuracy** across test cases through robust calibration and signal modelling<br>• Delivered a comprehensive software tool that effectively mirrors physical hardware behavior for user training |

## Positions of Responsibility

**President, Hall Executive Committee** | **Hall of Residence 5, IIT Kanpur** *(Jan'26-Present)*

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| **Leadership** | • Elected by **700-member electorate** to represent Hall 5 at institute level, including **CoSHA** & **HMC** bodies<br>• Led Hall 5 in **Inter-Hall Gen. Championship**, mobilizing **1500+ participants** across **Galaxy** & **Inferno** |
| **Initiatives** | • Led a **Mess Renovation (22 lakh)** and **Reading Room (5 lakh)** revamp to upgrade facilities and hygiene<br>• Collaborated with **Civil Engineering** profs on a **13.91 lakh rainwater harvesting** project for water reuse |
| **Impact** | • Partnered with **Agnys Waste Consultants** for waste composting, generating **INR 4k** monthly revenue overall<br>• Prevented layoffs of **15+ sanitation workers**, advocating for them among **250+ member Shramik Sanstha** |

## Technical Skills

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| **Languages** | • **Python, C, C++, Verilog, SQL, HTML/CSS**; strong grasp of algorithms, data structures, and systems |
| **Tools&Tech** | • **Django, React.js, NumPy, Scikit-Learn**; proficient in **Blender**, **Lightroom**, and **DaVinci** workflows |

## Relevant Coursework

***- Ongoing***

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| AI, ML & DL Applications (*) | Signals, Systems & Networks | Complex Variables |
| Fundamentals of Computing | Economy, society & Public Policy | Control Systems |
| Probability & Statistics | Linear Algebra | Partial Differential Equations |

## Extra-Curricular Activities

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| **Managerial** | • As **SnT Secretary**, led the major **20Cr+ Makerspace** hub launch, serving **10+ IITK clubs** and societies<br>• Sourced and curated **advanced equipment** (3D printers, laser cutters, PCB stations) for **Makerspace** hub |
| **Quant Research** | • Competed in the **Intl. Quant Championship** on **WorldQuant BRAIN**'s global alpha research platform<br>• Awarded **Research Consultant** role on **WorldQuant BRAIN** for strong **quantitative alpha** performance |
| **Lead** | • As **Galaxy'26 Hall Captain**, managed logistics & coordinated **1800+** participants across **55+** competitions<br>• As **SnT Secretary**, spearheaded the breakthrough **20Cr+ Makerspace** hub launch for **10+ IITK clubs** |